

Chapter 5: Mathematics Logic

Class 22 – Mental Mathematics, Strategy Games & Problem Solving

Learning Objectives

After completing this lesson, students will be able to:

- Perform mental calculations quickly and accurately.
 - Apply simple strategies for solving mathematical problems.
 - Improve speed and accuracy in basic arithmetic.
 - Develop logical thinking through strategy games.
 - Solve real-life mathematical problems confidently.
 - Strengthen decision-making and critical thinking skills.
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Introduction

Mental Mathematics helps students solve mathematical problems without using paper, calculators, or other tools. It improves concentration, memory, and confidence in calculations. Strategy games and logical puzzles make learning enjoyable while developing thinking and decision-making skills.

In this lesson, students will practice quick calculations, estimation, problem-solving, and fun mathematical games.

What is Mental Mathematics?

Mental Mathematics means solving mathematical problems in your mind without writing them down.

Benefits of Mental Mathematics

- Improves calculation speed.
- Develops memory.
- Builds confidence.
- Enhances concentration.

- Strengthens logical thinking.
 - Helps in everyday calculations.
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Mental Addition

Add numbers mentally using simple methods.

Examples

- $25 + 15 = 40$
 - $48 + 22 = 70$
 - $65 + 35 = 100$
 - $120 + 80 = 200$
 - $145 + 55 = 200$
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Mental Subtraction

Subtract numbers quickly in your mind.

Examples

- $60 - 25 = 35$
 - $100 - 45 = 55$
 - $150 - 70 = 80$
 - $200 - 90 = 110$
 - $500 - 150 = 350$
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Quick Multiplication

Multiply numbers mentally using simple patterns.

Examples

- $5 \times 6 = 30$
- $8 \times 7 = 56$
- $9 \times 10 = 90$
- $12 \times 5 = 60$

- $25 \times 4 = 100$
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Quick Division

Divide numbers mentally.

Examples

- $20 \div 5 = 4$
 - $36 \div 6 = 6$
 - $72 \div 8 = 9$
 - $100 \div 10 = 10$
 - $120 \div 12 = 10$
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Estimation Skills

Estimation means finding a number that is close to the exact answer.

Examples

- $49 \approx 50$
- $102 \approx 100$
- $198 \approx 200$
- $351 \approx 350$
- $799 \approx 800$

Uses of Estimation

- Shopping
 - Time management
 - Distance measurement
 - Budget planning
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Number Comparisons

Compare numbers using $>$, $<$, $=$

Examples

- $65 > 56$
 - $42 < 90$
 - $100 = 100$
 - $345 > 298$
 - $150 < 180$
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Time-Based Mental Mathematics

Solve simple time questions.

Examples

- 30 minutes + 30 minutes = **1 hour**
 - 2 hours + 1 hour = **3 hours**
 - Half of 60 minutes = **30 minutes**
 - Quarter of 1 hour = **15 minutes**
 - One day = **24 hours**
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Logical Problem Solving

Read the problem carefully and think step by step.

Examples

1. Riya has **15 apples**. She buys **10 more**. How many apples does she have?

Answer: 25 apples

2. A bus has **40 passengers**. **12 passengers** get down. How many passengers remain?

Answer: 28 passengers

3. A shopkeeper has **60 chocolates**. He sells **25 chocolates**. How many chocolates are left?

Answer: 35 chocolates

Sudoku (Basic)

Students complete a simple 4×4 **Sudoku** by filling in the missing numbers.

Rules

- Use numbers **1–4**.
 - Do not repeat a number in any row.
 - Do not repeat a number in any column.
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Magic Squares (Simple)

Arrange numbers so that each row and column has the same total.

Example

8 1 6
3 5 7
4 9 2

Each row, column, and diagonal adds up to **15**.

Strategy Games

Students play games that improve thinking and planning.

Examples

- Number Bingo
- Number Relay
- Domino Mathematics
- Puzzle Race
- Memory Card Game
- Number Maze

- Tangram Puzzle
 - Math Relay Challenge
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Decision-Making Activities

Students solve everyday mathematical situations.

Examples

- You have ₹100. You buy a notebook for ₹35 and a pencil for ₹15. How much money is left?

Answer: ₹50

- A school bus has **40 seats**. **35 students** are already seated. How many seats are empty?

Answer: 5 seats

Real-Life Mathematics Applications

Students observe how mathematics is used in daily life.

Examples

- Shopping
 - Banking
 - Cooking
 - Sports
 - Travel
 - Time Management
 - Measuring Distance
 - Counting Money
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Team Problem Solving

Students work in groups to solve mathematical challenges.

Sample Activities

- Complete number puzzles.
 - Solve word problems.
 - Arrange numbers correctly.
 - Find the missing number.
 - Explain the solution to the class.
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Revision

Students revise:

- Mental Addition
 - Mental Subtraction
 - Multiplication
 - Division
 - Estimation
 - Number Comparison
 - Time Problems
 - Sudoku
 - Strategy Games
 - Problem Solving
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Did You Know?

- Mental mathematics helps improve memory and concentration.
 - Sudoku puzzles strengthen logical thinking.
 - Strategy games improve planning and decision-making skills.
 - Mathematics is used every day in shopping, cooking, sports, and travel.
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Points to Remember

- Think before solving a problem.
- Use mental calculation whenever possible.
- Check your answers carefully.
- Read word problems slowly.
- Practice every day to improve speed.
- Work together and learn from others.